

Group: 3

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Water Filter Project



Materials: Sand, Pea Gravel, Drainage Rock, Cotton pad, Plastic Cup

During this project, we worked on multiple prototypes to design and implement a handheld sand-based water filter. The idea for this project is the make a cheap, easy to use, and effective water filter to be used around the globe in areas without access to clean water.

Day 1

The initial attempt contained a mostly even distribution of sand, pea gravel, and drainage rock.



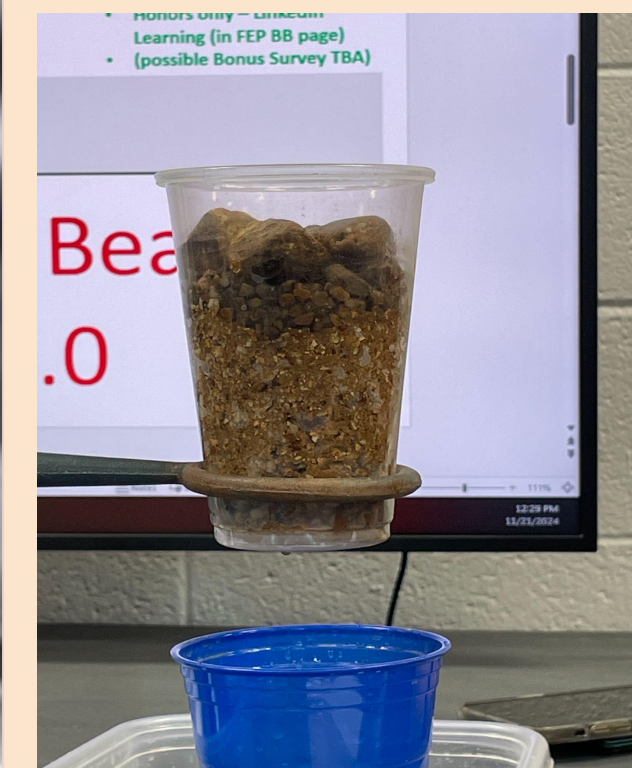
We used this prototype to decide on the right drainage method. We found 5 pin holes at the bottom of the cup most effective.

Day 2

This prototype was dedicated to analysing the effect sand had on transmittance. The amount of sand was increased dramatically, and took the majority of space in the cup. This increased the time it took for the water to properly filter, but the trade off seemed worth it due to the drastic increase in transmittance.

Day 3

After deciding that sand was the most effective filtration method, we experimented with layer design.



We added a second layer of pea gravel without increasing the total amount, and observed the effect on our transmittance. It was very positive.

This trend-line suggests that the more time spent filtering, the less effective each additional second becomes at removing particles. So for future projects, we need to find out the threshold of transmittance for safe drinking water, and minimize the amount of time it takes to get to that point.

Global Impact:

With further tweaking, optimization, and scaling, this design can help bring clean water out to rural areas. It can decrease the amount of time locals need to dedicate to their basic needs, and allow for them to invest in more advanced education/employment.

